

Phase 2 Sales Objection Fix Report

Reply Generation Pipeline — Vyapar AI Employee Platform

Status: FIXED PRODUCTION BUG

Date: June 2, 2026

Context: Sales mode activation working; Telegram replies still repeated package pitch

1. Problem Summary

Production logs confirmed the activation layer was fixed:

Log event	Status
LEAD_SIGNALS_DETECTED	Working
LEAD_UPSERTED	Working
SALES_MODE_ACTIVE	Working
EMPLOYEE_REPLY_GENERATED sales_mode=True	Working
Objection-aware replies	Broken — same template every turn

Telegram examples that failed

- discount chhaina?
- CGNET is cheaper / mahinaako 500 ma dinchha
- senior ko number dinus
- ma najodne mahango bhayo

The bot kept sending the same structure:

Khemraj Adhikari, tapai ko chaso ko lagi dhanyabad
→ 100/200 Mbps package + price
→ phone/WhatsApp number dinuhos
→ Dhanyawad!

2. Root Causes

2.1 No objection handling

There was zero logic for discount, competitor comparison, escalation, or rejection. Once `lead_stage=interested`, every message stayed in sales mode with the same pitch instructions.

2.2 Product block re-injected every turn

In `ai_employee_engine.py`, even with `sales_mode=True`, the system prompt always included a fresh `product_block` (package name + price). The model was re-fed the pitch on every turn.

2.3 SALES_EMPLOYEE_HINT replaced intent context

In `prompts.py`, when `sales_mode=True`, the specific intent hint was discarded and replaced by generic sales instructions — including “ask for phone if missing.”

2.4 Weak anti-repeat guidance

`sales_memory_to_prompt()` stored only the last Q/A (300 chars). Nothing explicitly told the model not to copy its previous pitch.

2.5 Single-turn OpenAI input

`openai_engine.py` sends one `user_prompt` per turn. All context must come from the system prompt. Without objection blocks, the model defaulted to the template.

3. Pipeline Trace (Before Fix)

```
User: "discount chhaina?"
→ lead exists, stage=interested, score=70
→ SALES_MODE_ACTIVE reason=lead_stage=interested ✓
→ product_block = "100 Mbps... NPR 10,000..." X re-pitched
→ intent_block = SALES_EMPLOYEE_HINT only X no discount handling
→ user_prompt = current message only
→ OpenAI → same package template again X
```

4. Fix Applied

4.1 New module: sales_objection.py

Detects four objection types and generates prompt hints:

Type	Example triggers	Behavior
discount	discount, chhut, offer	Answer discount question; no re-pitch
competitor	CGNET, Wlink, sasto chha	Acknowledge comparison; cite verified strengths
escalation	senior, manager, number dinus	Share official phone/toll-free from profile
rejection	najodne, mahango, chaso chhaina	Respect decline; do not ask for phone

4.2 prompts.py

- Added `ANTI_REPEAT_HINT` — do not copy previous package pitch
- `compose_system_prompt()` now accepts `objection_block` and `suppress_product_pitch`
- When objection detected, product pitch suppressed in system prompt

4.3 ai_employee_engine.py

- Wire `detect_sales_objection()` before prompt build
- Suppress `product_block` when objection detected
- Log `SALES_OBJECTION_DETECTED`
- Objection-specific rules in user prompt

4.4 lead_manager.py

Stronger anti-repeat instruction in `sales_memory_to_prompt()`.

4.5 tests/test_sales_objection.py

Unit tests for all four objection types from production examples.

5. Files Changed

File	Change
<code>sales_objection.py</code>	NEW — detection + prompt hints
<code>prompts.py</code>	Objection-aware compose + anti-repeat
<code>ai_employee_engine.py</code>	Wire objection pipeline + logging
<code>lead_manager.py</code>	Stronger sales memory anti-repeat
<code>tests/test_sales_objection.py</code>	NEW — unit tests

6. Expected Behavior After Fix

User message	Expected reply
discount chhaina?	Direct answer: no confirmed discount. No package re-pitch.
CGNET sasto chha	Acknowledge comparison, cite support/policy strengths. No template repeat.

senior ko number dinus	Share HONS phone/toll-free from company profile. No pitch.
ma najodne mahango bhayo	Respect decline. Do not ask for phone again.
mero naam ke ho?	Direct memory answer (unchanged)

7. Expected Logs

```
SALES_MODE_ACTIVE user_id=2071979418 company_id=hons reason=lead_stage=interested stage=interested score=70
SALES_OBJECTION_DETECTED user_id=2071979418 company_id=hons objection=discount suppress_pitch=True
EMPLOYEE_REPLY_GENERATED user_id=2071979418 company_id=hons sales_mode=True
```

8. Testing Checklist

1. Start sales conversation → confirm SALES_MODE_ACTIVE
2. Ask discount chhaina? → expect SALES_OBJECTION_DETECTED objection=discount
3. Say CGNET sasto chha → objection=competitor , no identical repeat
4. Say senior ko number dinus → objection=escalation , shows official contact
5. Say ma najodne mahango bhayo → objection=rejection , no phone ask
6. Ask mero naam ke ho? → direct memory answer

```
python -m unittest tests.test_sales_objection tests.test_sales_mode tests.test_lead_phase2 -v
```

9. Deploy

Deploy atomically on Render:

- sales_objection.py (new)
- ai_employee_engine.py
- prompts.py
- lead_manager.py

No new environment variables required.